

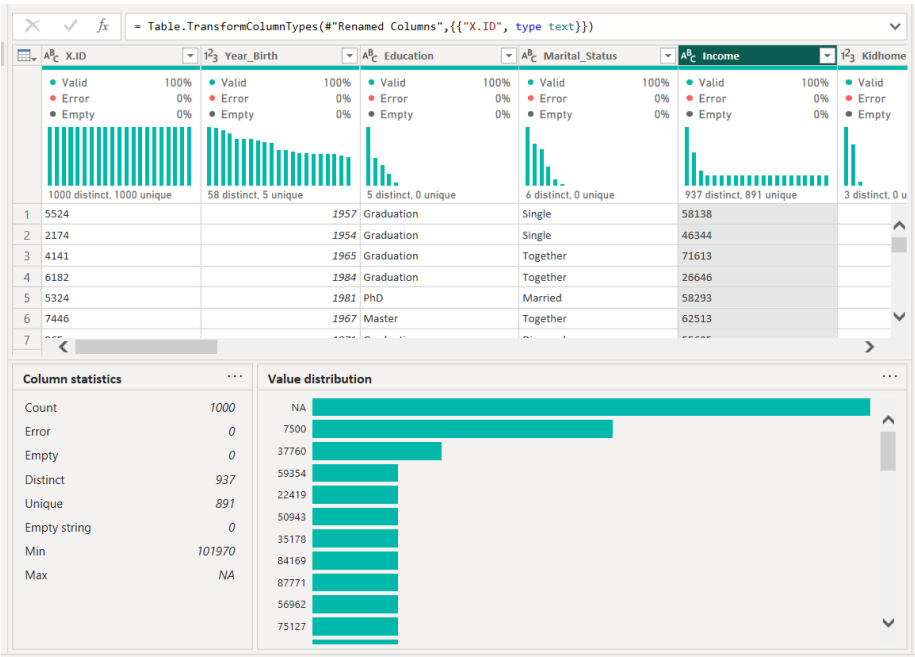
Marketing Campaign Data Preparation and Transformation

Integrating the data:

In this step, we imported the data from the CSV file into Power Query in import mode. This step ensures that the data is accessible and ready for the subsequent profiling and transformation steps.

Data Profiling:

Data profiling involves examining a dataset to gain an in-depth understanding of its characteristics, quality, structure, and content. This step is crucial in identifying any issues that could impact the analysis. During the data profiling of this dataset, it was found that some columns did not have the proper data types, such as the 'Income' column. Identifying these issues early allows for targeted data cleaning and transformation steps to ensure data integrity and reliability.



Data Transformation Steps:

Removing Leading Whitespace in Column Names

Problem: Some column names might have leading or trailing whitespace, causing inconsistency.

Solution: Removed leading and trailing whitespace from all column names.

Cleaning and Converting Response Column

Problem: The 'Response' column had non-numeric characters and was not in binary format.

Solution: Removed non-numeric characters “ ” and converted the column to binary format (0 or 1).

Handling Missing Values in 'Income'

Problem: Missing values in important columns could skew the analysis.

Solution: Imputed missing values in the 'Income' column with the median value because the distribution of the income is Right-skewed.

Handling Outliers in 'Year_Birth'

Problem: Outliers in the 'Year_Birth' column indicating unrealistically early years, such as 1893.

Solution: Removed rows with 'Year_Birth' less than 1920 to eliminate unrealistic data.

Handling Outliers in 'Income'

Problem: Extreme high values in the 'Income' column could skew analysis.

Solution: Capped 'Income' outliers at the 99th percentile to reduce skewness while retaining most data.

Cleaning Up 'Marital_Status'

Problem: The 'Marital_Status' column contained inconsistent or potentially incorrect categories such as 'Alone', 'Absurd', and 'YOLO'.

Solution:

- Standardized common categories ('Single', 'Together', 'Married', 'Divorced', 'Widow').
- Grouped less common categories ('Alone', 'Absurd', 'YOLO') into the “Single” category.

Converting of Data Types

Column Name	Data Type (Power Query)
X.ID	Text
Year_Birth	Whole Number
Education	Text
Marital_Status	Text
Income	Decimal Number
Kidhome	Whole Number
Teenhome	Whole Number
Dt_Customer	Date
Recency	Whole Number
MntWines	Whole Number
MntFruits	Whole Number
MntMeatProducts	Whole Number
MntFishProducts	Whole Number
MntSweetProducts	Whole Number
MntGoldProds	Whole Number
NumDealsPurchases	Whole Number
NumWebPurchases	Whole Number
NumCatalogPurchases	Whole Number
NumStorePurchases	Whole Number
NumWebVisitsMonth	Whole Number
AcceptedCmp3	Whole Number
AcceptedCmp4	Whole Number
AcceptedCmp5	Whole Number
AcceptedCmp1	Whole Number
AcceptedCmp2	Whole Number
Complain	Whole Number
Z_CostContact	Whole Number
Z_Revenue	Whole Number
Response	Whole Number

Enriching the data:

In this step I enhanced the dataset by adding new features or columns. To gimme more capabilities in data analysis and reporting.

Column Name	Description
ResponseToAnyCampaign	Indicates whether the customer has accepted any campaign (1 if they accepted any, 0 otherwise).
Age	The age of the customer, calculated from the birth year.
Customer_Generation	The generational cohort of the customer (e.g., Baby Boomer, Generation X, etc.) based on their birth year.
Dependents	The total number of dependents the customer has, calculated as the sum of Kidhome and Teenhome.
Has_children	A binary indicator (1 or 0) of whether the customer has any children.
IncomeLevel	A categorical representation of the customer's income level, typically derived from the Income column (e.g., Low, Medium, High).
Total_spent	The total amount of money the customer has spent, calculated as the sum of MntWines, MntFruits, MntMeatProducts, MntFishProducts, MntSweetProducts, and MntGoldProds.
Total_campaigns	The total number of campaigns the customer has been part of, calculated as the sum of AcceptedCmp1, AcceptedCmp2, AcceptedCmp3, AcceptedCmp4, and AcceptedCmp5.
Income (bins)	The income of the customer categorized into bins for analysis (e.g., 0-30k, 30-60k, etc.).
M_Score	The monetary score for the customer, part of the RFM (Recency, Frequency, Monetary) model, typically derived from the total spending amount.
R_Score	The recency score for the customer, part of the RFM model, typically derived from the Recency column.
Total Purchase	The total number of purchases made by the customer, calculated as the sum of NumDealsPurchases, NumWebPurchases, NumCatalogPurchases, and NumStorePurchases.

F_Score	The frequency score for the customer, part of the RFM model, typically derived from the total number of purchases.
RFM	The overall RFM score, a combination of R_Score, F_Score, and M_Score, used to segment customers based on their behavior.
Segment	The customer segment based on the RFM score or other segmentation criteria (e.g., loyal customers, new customers, etc.).
AgeGroup	The age group of the customer, typically categorized into bins (e.g., 20-30, 30-40, etc.).
TotalFoodExpenditure	The total amount spent on food products, calculated as the sum of MntWines, MntFruits, MntMeatProducts, MntFishProducts, and MntSweetProducts.

Data Modelling:

In this step I used the one big table approach because the dataset is small and don't have many dimensions to create a star or snowflake schema

X.ID	Year_Birth	Education	Marital_Status	Income	Kidhome	Teenhome	Dt_Customer	Recency	MntWines	MntFruits	MntMeatProducts	MntFishProducts	MntSweetProducts
9283	1978	Graduation	Single	60199	1	2	12/09/2013	49	8	1	7	2	
8737	1978	Graduation	Divorced	55563	0	1	05/04/2014	22	29	1	9	2	
10022	1973	PhD	Single	54466	1	1	09/02/2014	78	12	0	4	0	
10350	1950	PhD	Married	54432	2	1	09/05/2013	37	33	0	5	0	
5229	1969	PhD	Together	54132	0	1	16/02/2014	81	30	1	8	0	
5062	1963	PhD	Married	54072	1	1	27/07/2013	71	35	0	4	0	
8207	1957	2n Cycle	Married	53233	0	1	18/03/2014	18	28	0	9	3	
1055	1976	Master	Married	53204	1	1	20/03/2014	40	29	0	8	2	
1131	1967	PhD	Single	52569	0	1	05/04/2014	54	85	0	3	0	
7264	1978	2n Cycle	Single	52195	2	1	12/05/2014	2	12	0	4	0	
73	1953	PhD	Single	51411	1	2	26/05/2013	81	14	0	3	0	
9235	1957	Graduation	Single	51381	1	1	27/05/2014	45	7	0	8	2	
4345	1964	2n Cycle	Single	51381	1	1	12/01/2014	49	5	1	9	2	
3769	1972	PhD	Together	51381	1	0	02/03/2014	17	25	1	13	0	
10160	1956	Master	Together	50943	0	1	21/06/2013	49	31	0	7	2	
9347	1956	Master	Together	50943	0	1	21/06/2013	49	31	0	7	2	
6271	1952	PhD	Married	50870	0	1	19/06/2014	13	53	0	8	0	
2061	1955	Graduation	Married	50737	0	1	11/05/2013	61	78	0	11	0	
1158	1966	Graduation	Divorced	48877	0	1	05/01/2014	27	102	1	15	2	